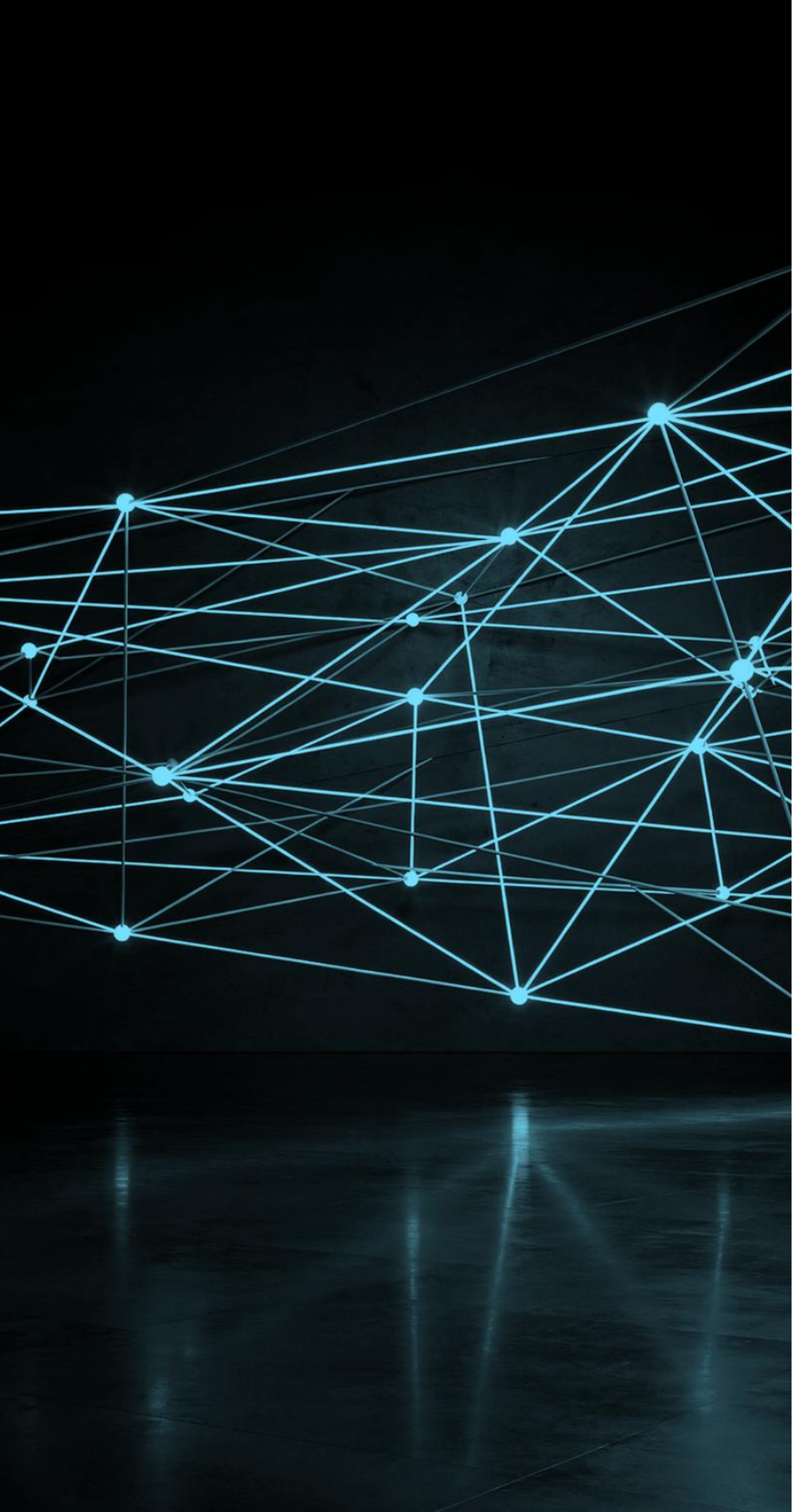


Ambient Systems

Integrated technical expertise in engineering, FP&A, AI, and IP.

Turning five years of deep-tech R&D into specialized consulting that delivers direct results.

Based from Belgrade with global operations in Hong Kong.

Introduction

Agency Overview

Driving financial performance through engineering, AI, and IP.

Background

Offering expertise after five years of building an award-winning, PCT-patented NUS deep-tech AI technology startup that achieved technical readiness (TRL 6).

Our Mandate

- ✓ Industrial applications
- ✓ Streamlined FP&A
- ✓ Intellectual Property R&D
- ✓ Software subscription (in development)

Service Lines:

1. **Civil Engineering & Energy Systems**
   Drafting and load calculations.
   Solar array design, microgrid optimization, building energy efficiency.
2. **Supply-Chain Management & Vendor Optimization**
   Sourcing strategy, customs optimization, vendor scorecards.
3. **Vertical Farming / CEA Optimization**
   Layout design, yield modeling, environmental controls.
4. **IP Strategy & Patent Support**
   Prior art searches, patent filing support, freedom-to-operate analysis.

Active on Leading Platforms: Upwork, Freelancer.com, Catalant, AlphaSights, and —connecting with clients who need immediate, specialized expertise.

Case Study

End-to-End Climate Management Solution for Indoor Farming

SITUATION or BUSINESS CHALLENGE

Indoor farmers face significant crop loss and high operational costs due to the inability to maintain optimal climate conditions. Existing solutions are fragmented, manual, and labor-intensive, often leading to unmonitored temperature and humidity imbalances that promote mold and pests, ultimately reducing yield quality and quantity.

APPROACH

- Defined the **full-stack ML solution** (sensors, optimization engine, dashboard) and the scalable B2B SaaS + Hardware business model.
- Secured pilot **Letters of Intent (LOIs)** with two key vertical farm operators to validate demand and establish the initial pipeline.
- Developed the financial model projecting a **24% improvement in client margins** with a rapid **4–6-month customer payback period**.
- Produced all investment deliverables (Investment Deck, Business Plan) and defined the IP/patent strategy for the core optimization algorithm.

OUTCOMES ACHIEVED

- 24% ROI Model: Developed a model projecting a 24% improvement in client margins with a rapid 4–6-month customer payback.
- Validation: Achieved critical market validation and established a pipeline through secured commercial LOIs.
- IP Foundation: Defined the patent strategy for the core optimization algorithm.

Previous Experience

Results achieved personally at prior ventures.

Case studies in distributed mining, SEC big pharma disclosures, and image recognition.

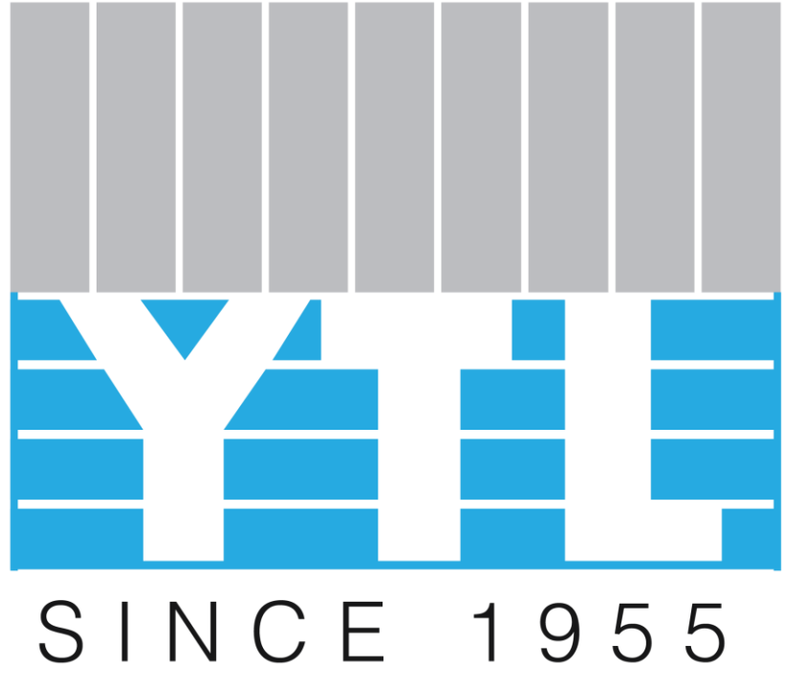
	Distributed Mining Company	Conflict Minerals	Image Neural Network
Challenge	Uncontrolled OpEx (Energy Costs). No link between real-time operational data and financial planning; resulting in reactive procurement.	Urgent SEC 3TG Conflict Minerals disclosure. Major pharmaceutical client required rapid compliance under a tight regulatory deadline.	Startup needed a robust, high-accuracy AI vision model to automatically detect artisanal mining by analyzing satellite imagery across complex biomes.
Action	Developed Big Data predictive OpEx models using operational and market data. Integrated models for real-time cost visibility at site level.	Led data analysis and consolidated raw data from 20+ international offices. Advised on compliance risk mitigation for SEC disclosure requirements.	Led as Principal Investigator (PI). Designed and implemented Convolutional Neural Network (CNN) models. Developed an automated data collection and prediction pipeline.
Result	12% sustained reduction in global energy costs, achieving millions in OpEx savings.	Successfully met critical U.S. regulatory disclosure. Delivery executed under urgent, one-day notice.	Delivered a validated CNN image classification model and system architecture.

Solution for YTL Corporation: Industrial Finance AI Assistant: Bridging Field Reporting and Cost Control (High-ROI PoC)

- **Challenge #1**: High-ROI digital transformation on construction sites.
- **Collaboration Need #4**: AI Assistants for Supervisors & Engineers

Delivery Milestones (4-week PoC Sprint):

- **Week 1: Foundations & Data Linkage**
  - Select one critical form for automation; Map data fields to the Ambient Platform's Financial Schema.
- **Week 2: AI Assistant Prototype**
  - Deploy Functional Online Prototype comparing known-data (Voice/Photo/text) to a structured checklist.
- **Week 3: Financial ROI Integration**
  - Real-Time Financial Risk Dashboard (linking field reports to).
- **Week 4: Validation & Testing**
  - Final ROI Analysis Report (Time/Cost Savings) vs. value-delivered (e.g., construction milestones met) against master engineer's drawings.



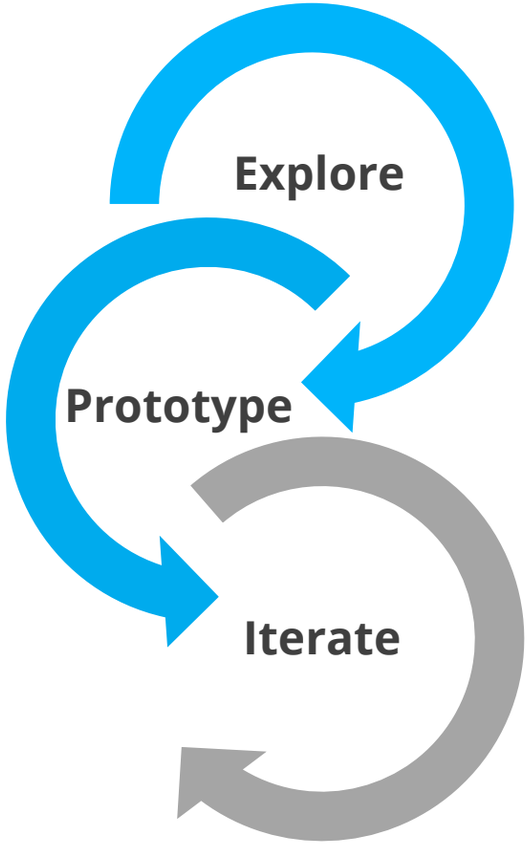
Potential Outcomes (ROI focused):

- **Burden** (Time Savings): fewer forms and easier accessibility to capture key data.
- **Bottleneck flagging:** lack of timely data, such as supervisor's notes or inventory changes, suggests that the progress is stuck when compared with best-practices sequence or the engineer's master drawings.
- **Financial Risk Attribution:** Immediate conversion of a reported defect (photo/text/voice) into a data point and remedy for a quicker ROI.

Business model

Our approach

Transparent Pricing



Book a Free 30-Minute Call

- We discuss your challenge, goals, and constraints.
- No pitch–just a questioning conversation to see if we can solve your problem.

Receive Fixed-Price Proposal in 48 Hours

- Clear scope, defined deliverables, flat fee.
- You know exactly what you're paying and what you're getting before committing a dollar.

Get Results in 30-60 Days

- Fast deliverable outcomes, tangible outcomes, and clear next steps.
- Most engagements pay for themselves in the first month.

Most engagements start between **€1,500–€3,500 fixed price**.
No hourly billing. No scope creep. No surprise invoices.

Ivan Damnjanovic
Founder
Technical Lead & Systems Architect

Entrepreneur and inventor from Belgrade with 10 years of experience across Asia (Singapore, Nepal, Hong Kong). Track record scaling YC-backed startup and cross-border intellectual property ventures. University of Toronto and NUS graduate in civil engineering and computer science.

Languages: Serbian & English (native); French (intermediate)

Let us bring the tech to you.

Current: Based in Belgrade with global operations in Hong Kong, available for remote engagements globally.
Future: Centralizing global operations in Belgrade after mid-2026 to serve European clients.

Languages: Serbian & English (native); French (intermediate)

Ready to Start?

Five years of deep-tech R&D evolved into affordable technical expertise and applications.

First call is free.
Quote in 24 hours.
Results in 4-6 weeks.

Book Your Calendar

Why Ambient Systems for Malaysia-Japan Fast Track Pitch

- Proven impact in global supply-chain and energy projects
  - 12 % OpEx reduction (decentralised mining client)
  - 24 % ROI (Singapore-based vertical farming project)
- Repeated collaboration with Japanese corporates and IP programmes
  - Panasonic, Google Cloud, NUS GRIP
- Expert team, Hong Kong incorporated, Singapore dedicated contact address
  - rapid 4-week delivery, fixed-price engagements
- Experienced in secure contracts, timely execution, and joint IP development
  - WIPO-certified processes
- Established network in Asia-Pacific and Europe
- Invoice from Hong Kong entity, Singapore mailing address provided
  - Immediately available for on-site or remote collaboration